Problem 1

Assume that the following code has been executed on a CPU with SCOREBOARD.

	Instruction	ISSUE	READ OPERAND	EXE COMPLETE	WB
I1	lw $1, 24($fp)	1	2	5	6
I2	lw $2, 8($fp)	2	3	6	7
I3	lw $3, 4($fp)	7	8	11	12
I4	add $2, $2, $3	8	13	16	17
I5	add $1, $1, $2	9	18	21	22
I6	sw $1, 16($fp)	10	23	26	27

A. List all the possible conflicts in the code.

B. Is there a “configuration” that can respect the shown execution?
How many units? Which kind? What latency?

C. If the previous table was not correct, please write the right one and specify the number, kind, and latency for each unit.

Answer 1.A

Answer 1.B

Yes, it exists. 2 LDU, 2 FPU (3cc latency for both kind of FUs)

Answer 1.C

	Instruction	ISSUE	READ OPERAND	EXE COMPLETE	WB
I1	lw $1, 24($fp)				
I2	lw $2, 8($fp)				
I3	lw $3, 4($fp)				
I4	add $2, $2, $3				
I5	add $1, $1, $2				
I6	sw $1, 16($fp)				

Problem 2

Please consider the program in the table to be executed on a CPU with dynamic scheduling based on TOMASULO. Consider an architecture with:

- 3 RESERVATION STATIONS (RS1, RS2, RS3) + 2 LDU units (LDU1, LDU2) with latency 3 cycles
- 2 RESERVATION STATIONS (RS4, RS5) + 2 FPU (FPU1, FPU2) with latency 3 cycles

1. Please complete the TOMASULO TABLE by assuming a cache MISS on I2 that adds 5 penalty cycles.

	Instruction	ISSUE	Start EXE	WB
I1	ld $1, 28($fp)	1	2	5
I2	ld $2, 16($fp)	2	3	11
I3	addd $1, $1, $2	3	12	15
I4	ld $2, 12($fp)	4	6	9
I5	addd $2, $1, $2	5	16	19
I6	addd $1, $1, 1	16	17	20
I7	sd $1, 0($2)	17	21	24

Problem 3

Describe (the answer has to be effectively supported) a 1-BHT and a 2-BHT able to execute the following assembly code (R0 is set to 4, R1 is set to 0)

```
LOOP:    LD     F1   0     R0
         ADDD   F2   F1    F1
         DIVD   F3   F2    F1
         ADDI   R1   R1    1000
LOOP2:   MULTD  F12  F3    F2
         SUBI   R1   R1    1
         BNEZ   R1   LOOP2
         SUBI   R0   R0    4
         BNEZ   R0   LOOP
```

The obtained result, in terms of miss predictions, is inline with theoretical characteristics of the two predictors? Please effectively support your answer.

Answer